TRAUMA · NEET-SS / FET / INI-SS

Hip and femoral fractures

1. Applied anatomy of the proximal femur

- **Neck-shaft angle** about 125-135 degrees; **anteversion** about 10-15 degrees in adults.
- **Blood supply of the femoral head:** mainly the **lateral epiphyseal (retinacular) branches of the medial femoral circumflex artery (MFCA)**, running up the posterosuperior neck. Lesser contributions: inferior gluteal, LFCA (anterior neck), and the artery of the ligamentum teres (small in adults; important in children). **Intracapsular fractures** disrupt the retinacular vessels: **AVN and non-union**.
- **Calcar femorale:** dense posteromedial cortical bone (load-bearing); **Ward's triangle** is the weak area between trabecular groups (Singh index grades osteoporosis from trabecular patterns).
- The **intertrochanteric line and capsule:** the capsule attaches anteriorly to the intertrochanteric line, but posteriorly only halfway down the neck.

2. Femoral neck fractures

Classifications

Garden	Description	Practical grouping
I	Incomplete / **valgus impacted**	Undisplaced
II	Complete, undisplaced	Undisplaced
III	Complete, **partially displaced** (trabeculae misaligned)	Displaced

Garden	Description	Practical grouping
IV	Complete, **fully displaced** (trabeculae of head realign with the acetabulum)	Displaced

- **Pauwels** (angle of the fracture line from horizontal; for young, high-energy fractures): **I under 30 degrees**, **II 30-50 degrees**, **III over 50 degrees (vertical shear, highest non-union and fixation failure)**.
- Location: subcapital, transcervical, basicervical (behaves like an extracapsular fracture).

Elderly (low-energy)

- **Undisplaced (Garden I-II): cannulated screws** (3 parallel screws in an inverted triangle, the inferior screw on the calcar, posterior screw against the posterior cortex; no screw entry below the lesser trochanter level: subtrochanteric fracture risk) or a sliding hip screw. Some evidence favours arthroplasty in very elderly frail patients with posterior tilt over about 20 degrees (high failure).
- **Displaced (Garden III-IV): arthroplasty**:
  - **Total hip arthroplasty** for active, independent, cognitively intact patients (better function; **HEALTH trial** showed small functional gains with more dislocations; NICE recommends THA for those who walk independently outdoors with a stick or less, are not cognitively impaired and are medically fit).
  - **Hemiarthroplasty** (bipolar or unipolar) for frailer, less active patients. **Cemented stems** are recommended (less pain, fewer periprosthetic fractures, better function: **WHiTE 5**), with care for **bone cement implantation syndrome**.
- **Orthogeriatric care:** surgery **within 36-48 hours** (NICE: on the day of or day after admission), early mobilisation, delirium prevention, nutrition, bone health (**secondary fracture prevention**:

bisphosphonate, vitamin D), falls assessment. **One-year mortality about 25-30%.**

Young (high-energy)

- An **emergency**: **urgent anatomical reduction** (closed or open, Watson-Jones or Smith-Petersen approach) and **stable fixation** (cannulated screws, sliding hip screw ± derotation screw, or a fixed-angle device such as a **dynamic hip screw with anti-rotation screw** or **proximal femoral locking/medial buttress plate** for vertical Pauwels III fractures).
- Complications: **AVN (about 20-30%)**, **non-union (about 10-30%)** (valgus intertrochanteric osteotomy (**Pauwels osteotomy**) converts shear into compression for non-union in young patients).
- **Ipsilateral femoral neck and shaft fracture:** the neck fracture (vertical, basicervical, undisplaced) is **missed in about 20-30%**: get a dedicated CT/fine-cut through the neck. Prioritise the **neck** (anatomical reduction).

Stress fractures of the femoral neck (Fullerton/Devas)

- **Compression (inferomedial)**: protected weight bearing; fix if the line exceeds about 50% of the neck or does not heal.
- **Tension (superolateral)**: high risk of displacement: **fix with screws**. Female athlete triad / RED-S.

3. Intertrochanteric (pertrochanteric) fractures

- Extracapsular; good blood supply, **unions reliably**; problem is fixation failure (**cut-out**).
- **Classification (AO/OTA 31A):**
  - **A1:** simple pertrochanteric (two-part; stable; intact lateral wall).
  - **A2:** multifragmentary pertrochanteric (posteromedial comminution, lateral wall incompetent in A2.2-2.3 per 2018).
  - **A3: intertrochanteric/reverse oblique** (unstable).

  - Evans (stable vs unstable; reverse obliquity is unstable), Boyd and Griffin, Jensen.
- **Unstable features: posteromedial comminution** (lesser trochanter), **reverse obliquity**, **subtrochanteric extension**, **lateral wall fracture** (thickness under about 20.5 mm predicts failure; Hsu), and inability to reduce.
- **Implants:**
  - **Sliding (dynamic) hip screw (DHS)** for **stable** fractures (A1): controlled collapse along the barrel.
  - **Cephalomedullary nail** (PFN, PFNA, Gamma, TFN, InterTAN) for **unstable** fractures, **reverse oblique** and subtrochanteric extension, deficient lateral wall. **DHS is contraindicated in reverse obliquity** (the fracture slides medially along the lag screw axis).
- **Tip-apex distance (TAD, Baumgaertner):** sum of the distances from the lag screw tip to the head apex on AP and lateral views, corrected for magnification. **Keep under 25 mm** (cut-out risk rises sharply above 25 mm). Position **centre-centre or inferior-centre** on AP, centre on lateral.
- **Complications: cut-out** (commonest mechanical failure; poor TAD, superior/anterior screw placement, unstable pattern), **Z-effect** (two-screw nails), **varus collapse**, periprosthetic fracture at the tip of a short nail, non-union (rare), malrotation.

4. Subtrochanteric fractures

- Within **5 cm distal to the lesser trochanter**. **High stress concentration** (medial compression, lateral tension).
- Deforming forces on the proximal fragment: **flexion (iliopsoas), abduction (gluteus medius/minimus) and external rotation (short external rotators)**; the shaft is pulled into **adduction (adductors)** and shortening.

- **Russell-Taylor** (piriformis fossa and lesser trochanter involvement); Seinsheimer.
- **Treatment: cephalomedullary (reconstruction) nail**, often after **open or clamp-assisted reduction** (reduction is the key; a malreduced varus fracture fails). Plates (95-degree blade plate, locking plates) in selected cases.
- **Atypical femoral fractures (AFF) with long-term bisphosphonates (ASBMR criteria):** subtrochanteric or diaphyseal, **minimal or no trauma**, **transverse or short oblique** with a **medial spike** when complete, **non-comminuted**, **lateral cortical "beaking"/thickening**, prodromal thigh pain, often **bilateral** (image the other femur). **Stop the bisphosphonate**, calcium/vitamin D, consider **teriparatide**; **prophylactic intramedullary nailing** for incomplete fractures with pain or a visible fracture line.

5. Femoral shaft fractures

- High-energy injuries; blood loss up to about 1-1.5 litres; **fat embolism** risk; associated injuries (ipsilateral neck fracture, knee ligament injuries, hip dislocation).
- **Winquist-Hansen** comminution grades 0-IV.
- **Treatment of choice: reamed, statically locked antegrade intramedullary nail** within 24 hours in a stable patient. **Early fixation reduces pulmonary complications** (ARDS, fat embolism) and allows mobilisation.
- **Antegrade nail entry: piriformis fossa** (straight nails) or **trochanteric** entry (curved-proximal nails; avoids iatrogenic neck damage; small abductor damage). **Retrograde nail** (through the intercondylar notch): obesity, bilateral fractures, **ipsilateral femoral neck fracture** (with separate neck fixation), floating knee, pregnancy, distal fractures, ipsilateral acetabular fracture.
- **Reaming** increases union rates (**SPRINT for tibia** showed no harm; for femur, reamed nails union better); theoretical fat embolism risk.

- **Damage control orthopaedics (DCO): external fixation first**, nail later (within about 2 weeks; ideally day 4-14 when the inflammatory "second hit" settles), in **borderline or unstable** polytrauma patients (see polytrauma notes).
- **Complications:**
  - **Malrotation** (commonest malalignment after nailing; compare with the other side; over about 15 degrees is significant).
  - **Limb-length discrepancy**, **non-union** (about 1-5%; exchange nailing for aseptic hypertrophic non-union), malunion (valgus with proximal fractures, apex-anterior with distal entry).
  - **Fat embolism syndrome:** 24-72 hours, **Gurd and Wilson criteria** (major: respiratory insufficiency, cerebral signs, **petechial rash** (axilla, conjunctiva)); supportive care.
  - Pudendal nerve palsy (traction table perineal post), heterotopic ossification at the entry, knee pain (retrograde).

6. Distal femur fractures

- **AO/OTA 33A (extra-articular), B (partial articular; B3 = coronal "Hoffa" fracture), C (complete articular).**
- **Hoffa fracture:** coronal plane fracture of a femoral condyle (**lateral more common**); seen only on lateral views/CT; fix with **anteroposterior lag screws** (perpendicular to the fracture).
- **Treatment: lateral locking plate** (MIPO) or **retrograde nail** (extra-articular and simple articular); **dual plating (lateral + medial)** for comminuted, bone loss or osteoporotic fractures and in nonunions; **distal femoral replacement** in elderly osteoporotic patients with very distal or periprosthetic fractures.
- **Deforming forces:** the **gastrocnemius** pulls the distal fragment into **extension (apex posterior)**; quadriceps and hamstrings shorten.
- **Periprosthetic distal femur fracture above a TKA:** Lewis-Rorabeck (I undisplaced, II displaced with a stable prosthesis, III loose

prosthesis) or Su classification; fix (plate, retrograde nail if the box allows) or revise/replace if loose.

7. Patella (often with knee trauma)

- Undisplaced with intact extensor mechanism (can straight-leg raise): cylinder cast/brace.
- **Displaced (step-off over 2 mm or gap over 3 mm) or extensor lag: tension-band wiring** (K-wires + figure-of-eight; converts tension to compression) or cannulated screws with tension band; partial patellectomy for comminuted inferior pole.
- **Bipartite patella:** superolateral (Saupe type III), bilateral, smooth rounded margins: not a fracture.

> Exam pearl: **Garden III-IV in the elderly = arthroplasty (cemented; THA if active and cognitively intact). Young displaced neck fracture = emergency anatomical reduction. Pauwels III = vertical shear, highest failure.**

> Exam pearl: **TAD under 25 mm prevents cut-out. DHS for stable; nail for unstable, reverse oblique and subtrochanteric. Atypical femoral fracture: lateral cortical beaking, transverse, bilateral: stop bisphosphonates, consider teriparatide, nail.**

> Exam pearl: **Ipsilateral neck + shaft fracture: the neck is often missed; prioritise the neck.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.